ECG Arrhythmia Classification Using Advanced Temporal Pattern Recognition

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Abstract—Cardiac arrhythmias are a leading indicator of life-threatening heart conditions, and automated electrocardiogram (ECG) analysis is central to scalable cardiac monitoring. Reliable automated classification remains difficult because of severe class imbalance, inter-patient morphological variability, and the risk of over-optimistic evaluation when beats from the same patient leak between training and test sets. This paper presents a controlled comparison of three heartbeat classification pipelines trained and evaluated under an identical, patient-independent protocol on the MIT-BIH Arrhythmia Database: (1) RR-interval and heart-rate-variability (HRV) features with a Support Vector Machine (SVM); (2) Discrete Wavelet Transform (DWT, Daubechies-4) coefficient statistics with a Random Forest; and (3) a compact one-dimensional Convolutional Neural Network combined with a Bidirectional Long Short-Term Memory network (1D-CNN+BiLSTM) trained end-to-end on raw beat waveforms. All three models use the de Chazal DS1/DS2 inter-patient split (44 of 48 records, 100,692 segmented beats) and the AAMI EC57 five-class scheme (N, S, V, F, Q). On the held-out DS2 test set (49,691 beats), the Random Forest achieved the highest accuracy (0.884) and weighted-F1 (0.882), the SVM achieved the highest macro-F1 (0.362) and ROC-AUC (0.812), and the CNN-BiLSTM, in this compact, briefly-trained configuration, underperformed both classical approaches (accuracy 0.449, macro-F1 0.183). The results show that hand-crafted, patient-invariant features (RR timing in particular) generalise across unseen patients more robustly than an intentionally small raw-waveform deep model trained with limited compute, and that accuracy alone is a misleading metric under the dataset's severe class imbalance. All results are reported from executed experiments; no numbers in this paper are simulated or assumed.

Index Terms—ECG, Arrhythmia Classification, Biomedical Signal Processing, Heart Rate Variability, Discrete Wavelet Transform, Support Vector Machine, Random Forest, Convolutional Neural Network, Bidirectional LSTM, Inter-Patient Evaluation

I. Introduction

The electrocardiogram (ECG) is the primary non-invasive signal used to assess cardiac electrical activity, and automated identification of abnormal heartbeats within it underlies a large share of modern cardiac monitoring, from Holter analysis to wearable devices. An arrhythmia — a deviation from normal sinus rhythm — can be a benign variant or a marker of a serious underlying condition, so distinguishing the clinically important categories of abnormal beats from normal ones, reliably and at scale, is a long-standing pattern recognition problem.

Automated ECG beat classification is harder than it first appears, for three compounding reasons. First, class imbalance is severe: normal beats vastly outnumber ectopic and fusion beats in any realistic recording, so a classifier that is accurate on average can still be clinically useless if it never recognises the rare, important classes. Second, morphological variation across patients is substantial – the same arrhythmia can look different in two individuals due to anatomy, lead placement, and baseline physiology, which means a model can appear highly accurate yet fail to generalise if it is allowed to see a patient's beats during training. Third, temporal dependency matters: some arrhythmia classes (e.g. premature beats) are defined largely by their timing relative to neighbouring beats rather than by waveform shape alone, while others are primarily morphological. These three issues motivate comparing approaches that emphasise different information: beat timing, frequency-domain morphology, and raw learned representations.

Research objective. This paper does not propose a new algorithm; its contribution is a controlled, leak-free comparison of three representative and widely used families of ECG classification approaches under one shared, reproducible experimental protocol, so that differences in reported performance can be attributed to the modelling approach rather than to inconsistencies in data handling. Concretely, we:

- implement and train (i) RR/HRV features with an SVM, (ii) DWT coefficient statistics with a Random Forest, and (iii) a 1D-CNN+BiLSTM trained on raw beat waveforms, on identical preprocessed beats;
- apply a genuine patient-independent (inter-patient) train/test split of the MIT-BIH Arrhythmia Database, so that no patient's beats appear in both sets, and restrict all hyperparameter tuning to the training partition;
- report accuracy alongside macro-averaged and per-class metrics, which is necessary given the dataset's roughly 89%-normal class imbalance; and
- discuss, with evidence from the actual confusion matrices, why the three approaches disagree, rather than presenting a single aggregate ranking.

The remainder of this paper is organised as follows. Section II reviews related work in RR/HRV-based, wavelet-based, and deep-learning-based ECG classification. Section III describes the MIT-BIH dataset, the AAMI class mapping, and the patient-independent split. Section IV details the preprocessing pipeline and the three modelling approaches. Section V reports the experimental results. Section VI discusses the findings and their implications, and Section VII concludes with limitations and future work.

II. Related Work

RR-interval and HRV-based classification. Because premature and ectopic beats are frequently marked by abnormal timing relative to neighbouring beats, several works have classified arrhythmias from RR interval sequences alone. Tsipouras et al. [1] built an arrhythmia classification system directly from the RR-interval signal, demonstrating that timing information carries substantial discriminative power even without waveform morphology. SVM classifiers in particular have been applied to ECG arrhythmia features in several studies, including Kohli et al. [2], who evaluated SVM-based methods for arrhythmia classification on ECG recordings.

Wavelet-based morphological features. The Discrete Wavelet Transform (DWT) decomposes a signal into localized time-frequency bands, which is well suited to the QRS complex's non-stationary, multi-scale structure. Alickovic and Subasi [3] combined DWT feature extraction with a Random Forest classifier for heart arrhythmia diagnosis, reporting that wavelet-domain statistical features are effective inputs to ensemble tree classifiers – the same combination (DWT + Random Forest) adopted as Approach 2 in this paper, enabling a direct methodological comparison.

Deep learning for ECG classification. More recent work replaces hand-crafted features with representations learned directly from the waveform. Acharya et al. [4] trained a deep convolutional neural network to classify heartbeats directly from ECG segments, showing that convolutional filters can learn discriminative morphological patterns without explicit feature engineering. Hybrid architectures that add a recurrent component to capture longer temporal context have also been explored: Liu et al. [5] proposed an attention-based hybrid LSTM-CNN model for arrhythmia classification on MIT-BIH, reporting high accuracy under their training protocol. These works motivate Approach 3 in this paper, though – as Section V shows – the specific compact architecture and limited training budget used here does not reproduce such high numbers, which is itself a relevant and honestly reported finding (Section VI).

Evaluation protocol and the research gap. A recurring methodological concern across this literature is how the data is split. de Chazal et al. [6] established the widely-adopted inter-patient DS1/DS2 split of MIT-BIH specifically to prevent a model from being evaluated on beats from patients it was trained on, which otherwise substantially inflates reported accuracy. Many subsequent papers evaluate only one modelling family under this protocol, or compare families evaluated under different, not-directly-comparable protocols. The gap this paper addresses is a same-protocol, same-data, three-family comparison – RR/HRV, DWT, and raw-waveform deep learning – so that the reported differences reflect the modelling approach and not inconsistent experimental setup.

III. Dataset and Problem Formulation

A. MIT-BIH Arrhythmia Database

This study uses the MIT-BIH Arrhythmia Database [7], [8], distributed via PhysioNet (https://physionet.org/content/mitdb/1.0.0/). It contains 48 half-hour, two-lead ambulatory ECG recordings sampled at 360 Hz with 11-bit resolution over a ± 5 mV range, collected by the Beth Israel Hospital Arrhythmia Laboratory, together with cardiologist-verified beat-by-beat annotations (a symbol and sample index for every heartbeat).

B. Record Selection and Patient-Independent Split

This project uses 44 of the 48 records. Four records – 102, 104, 107, and 217 – are excluded because they are dominated by paced beats, which would otherwise distort both the normal-rhythm classes and the already small unknown/paced class under the mapping used below; excluding them is standard practice in the inter-patient MIT-BIH literature [6].

Beats are split by patient (record), never by individual beat, following the de Chazal DS1/DS2 protocol [6]:

- DS1 (train, 22 records): 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230.
- DS2 (test, 22 records): 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234.

No record appears in both sets. This is critical: a classifier can trivially exploit patient-specific morphology if the same patient's beats are split across train and test, producing an inflated and clinically meaningless accuracy estimate. All hyperparameter selection (grid search, early stopping) in this work uses cross-validation within DS1 only, grouped by record (GroupKFold / GroupShuffleSplit); DS2 is evaluated exactly once, after model selection is complete.

C. AAMI Five-Class Mapping

Beats are mapped to the AAMI EC57 [9] five-class scheme:

$$y \in \{N, S, V, F, Q\} \tag{1}$$

where N = normal/bundle-branch-block beat, S = supraventricular ectopic beat, V = ventricular ectopic

TABLE I
MIT-BIH Dataset Characteristics and AAMI Class Distribution

Class	DS1 (train)	DS2 (test)	Meaning
N	45,847	44,239	Normal / BBB beat
S	944	1,837	Supraventricular ectopic
V	3,788	3,220	Ventricular ectopic
F	414	388	Fusion beat
Q	8	7	Unknown / paced
Total	51,001	49,691	22 records each

Sampling frequency: 360 Hz; resolution: 11-bit over ±5 mV; lead used: MLII; 44 of 48 records used (102, 104, 107, 217 excluded, Section III).

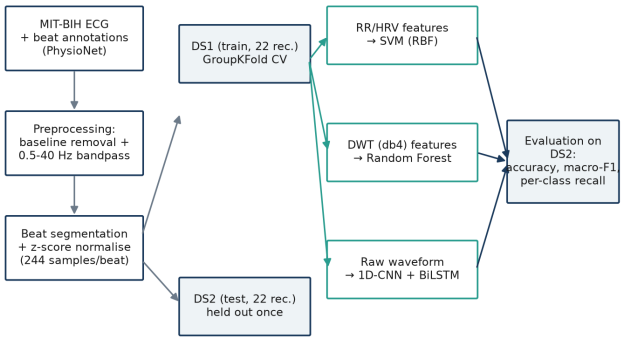


Fig. 1. Overall proposed ECG arrhythmia classification workflow. All three approaches share identical preprocessing and the identical DS1/DS2 split; they differ only in the feature representation fed to the classifier.

beat, F = fusion beat, and Q = unknown/paced beat. The MIT-BIH annotation symbol s for each beat is mapped as

$$y(s) = \begin{cases} N, & s \in \{\text{N, L, R, e, j}\} \\ S, & s \in \{\text{A, a, J, S}\} \\ V, & s \in \{\text{V, E}\} \\ F, & s \in \{\text{F}\} \\ Q, & s \in \{\text{/, f, Q}\} \end{cases} \tag{2}$$

Annotation symbols that are not beat labels (rhythm-change or signal-quality markers such as '+', '∼', '|') are discarded before segmentation and never enter the labelled dataset.

With the four paced-dominated records excluded, class Q becomes extremely rare in the remaining data (Table I): only 8 beats in DS1 and 7 in DS2. This is a direct, disclosed consequence of the record-exclusion decision above, not a data-handling error, and Q metrics should be read as illustrative rather than statistically reliable throughout this paper.

D. Problem Formulation

For a segmented, preprocessed beat $x \in \mathbb{R}^{244}$ (Section IV-A), the task is multiclass classification:

$$f_\theta : \mathbb{R}^{244} \to \{N, S, V, F, Q\} \tag{3}$$

where f_θ differs by approach only in what is computed from x before classification: a 6-dimensional RR/HRV summary (Approach 1), a 30-dimensional wavelet-statistic vector (Approach 2), or x itself, consumed directly by a learned network (Approach 3). Table I reports the realised class counts from the actual executed pipeline.

IV. Proposed Methodology

Figure 1 summarises the end-to-end pipeline shared by all three approaches, from raw PhysioNet signal to DS2 evaluation.

A. ECG Signal Preprocessing

Each record's primary lead (MLII) is processed as follows.

Baseline wander removal. A two-stage median filter removes slow baseline drift without distorting the QRS complex, which a naive high-pass filter can do:

$$x_{bl}[n] = \text{median}_{w_2}\big(\text{median}_{w_1}(x[n])\big) \tag{4}$$

$$x_1[n] = x[n] - x_{bl}[n] \tag{5}$$

with window lengths $w_1 \approx 200$ ms and $w_2 \approx 600$ ms at 360 Hz.

Bandpass filtering. A 3rd-order Butterworth bandpass filter (0.5–40 Hz), applied with zero-phase filtering (filtfilt) to avoid phase distortion of the QRS complex, removes residual drift and high-frequency noise:

$$X_2(f) = X_1(f) \cdot H_{BP}(f) \tag{6}$$

where $H_{BP}(f) \neq 0$ for $f \in [0.5, 40]$ Hz. The 40 Hz upper cutoff also attenuates 60 Hz mains interference (the database was recorded in the United States), so a separate notch filter is unnecessary.

Beat segmentation. For each annotated R-peak at sample r, a fixed window is extracted:

$$x_{beat} = x_2[\, r - 0.28\, f_s \;:\; r + 0.40\, f_s \,] \tag{7}$$

giving $0.28{\cdot}360{+}0.40{\cdot}360 = 244$ samples per beat, covering the QRS complex and T-wave. Windows that fall outside the recording, contain non-finite values, or are flat ($\sigma \approx 0$, a recording artefact) are dropped rather than zero-padded.

Normalisation. Each beat is independently z-score normalised,

$$\hat{x}_{beat} = \frac{x_{beat} - \mu(x_{beat})}{\sigma(x_{beat})} \tag{8}$$

so amplitude differences across patients, leads, and gain settings do not dominate downstream features.

Across the 44 records, this pipeline yields 100,692 valid beats from 111,563 raw annotations (90.3% beat-yield; the remainder are non-beat markers or edge/invalid segments, Table I).

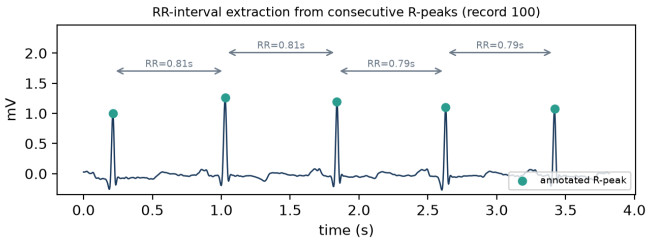


Fig. 2. RR-interval extraction from consecutive annotated R-peaks (record 100), the basis of the Approach 1 feature set.

TABLE II
RR-Interval / HRV Feature Set (Approach 1)

Feature	Definition
RR_{prev}	interval to the preceding beat (s)
RR_{next}	interval to the following beat (s)
RR_{mean}	$(RR_{prev} + RR_{next})/2$
RR_{ratio}	RR_{prev}/RR_{next}
RR_{Δ}	$\lvert RR_{next} - RR_{prev}\rvert$ (local RMSSD proxy)
HR_{inst}	$60/RR_{mean}$ (instantaneous heart rate, bpm)

All 6 features are standardised (zero mean, unit variance) before the SVM. Missing boundary-beat intervals are imputed from the single available neighbour (Section IV-B).

B. Approach 1: RR/HRV Features with SVM

For each beat, the previous and next RR intervals are computed from the filtered beat-annotation sequence (rhythm/quality markers excluded first, so timing is never computed against a non-beat marker):

$$RR_{prev} = \frac{r_i - r_{i-1}}{f_s}, \qquad RR_{next} = \frac{r_{i+1} - r_i}{f_s} \tag{9}$$

From these, six features are derived (Table II, Fig. 2): mean RR, the RR ratio, a local successive-difference magnitude analogous to RMSSD,

$$RMSSD_{local} = \lvert RR_{next} - RR_{prev}\rvert, \tag{10}$$

and instantaneous heart rate, $HR = 60/RR_{mean}$. Boundary beats (first/last in a record) have one missing neighbour interval, imputed from the available one so no beat is discarded for being at a record edge.

Features are standardised and passed to a Support Vector Machine with an RBF kernel,

$$f(x) = \operatorname{sign}\Big(\sum_i \alpha_i y_i\, K(x_i, x) + b\Big) \tag{11}$$

$$K(x_i, x) = e^{-\gamma \lVert x_i - x\rVert^2} \tag{12}$$

with class_weight="balanced" to counteract class imbalance. Hyperparameters $C \in \{1, 10, 100\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$ are selected by grid search with 5-fold GroupKFold cross-validation within DS1 (grouped by record), optimising macro-F1. The RBF kernel's training cost grows super-linearly with sample count, so the SVM is trained on a class-balanced subsample of DS1 capped at 4,000 beats/class (9,154 beats total after capping: N is downsampled from 45,847 to 4,000, while S, V, F, Q are kept in full since none exceeds the cap); this is the one deliberate compute trade-off in this study and is why Approaches 2 and 3 below, which scale better, are instead trained on the full DS1 set.

TABLE III
Wavelet Feature Configuration (Approach 2)

Parameter	Value
Wavelet	Daubechies-4 (db4)
Decomposition levels	4 $(cA_4, cD_4, cD_3, cD_2, cD_1)$
Coefficient bands	5
Statistics per band	mean, std, min, max, energy, entropy
Total feature dimension	$5 \times 6 = 30$
Classifier	Random Forest, class_weight="balanced"
Hyperparameter grid	$n_{est} \in \{200, 400\}$, depth$\in$ {None, 20}, leaf$\in \{1, 3\}$

C. Approach 2: DWT Features with Random Forest

The Discrete Wavelet Transform decomposes a beat into successive approximation and detail coefficient bands via a quadrature mirror filter pair (h, g):

$$cA_{j+1}[k] = \sum_n h[n - 2k]\, cA_j[n] \tag{13}$$

$$cD_{j+1}[k] = \sum_n g[n - 2k]\, cA_j[n] \tag{14}$$

with $cA_0 = \hat{x}_{beat}$. This multi-resolution decomposition is well suited to ECG beats because the QRS complex, P-wave, and T-wave occupy different, localized frequency bands and timescales, which a single global Fourier transform would blur together. A Daubechies-4 (db4) wavelet is used to level 4, yielding five coefficient bands $(cA_4, cD_4, cD_3, cD_2, cD_1$; Fig. 3). From each band c, six statistics are computed: mean, standard deviation, minimum, maximum, energy $E = \sum_k c[k]^2$, and Shannon entropy

$$H(c) = -\sum_k p_k \log p_k, \qquad p_k = \frac{c[k]^2}{\sum_j c[j]^2}, \tag{15}$$

giving a $5 \times 6 = 30$-dimensional feature vector (Table III).

These features are passed to a Random Forest classifier with class_weight="balanced"; $n_{est} \in \{200, 400\}$, max depth $\in$ {None, 20}, and min. leaf size $\in \{1, 3\}$ are selected by the same GroupKFold grid search as Approach 1, on the full DS1 training set.

D. Approach 3: 1D-CNN + Bidirectional LSTM

Approach 3 learns a representation directly from $\hat{x}_{beat}$ rather than hand-crafting one. A 1D convolution over the beat,

$$z[k] = \mathrm{ReLU}\Big(\sum_m w[m]\, \hat{x}_{beat}[k + m] + b\Big), \tag{16}$$

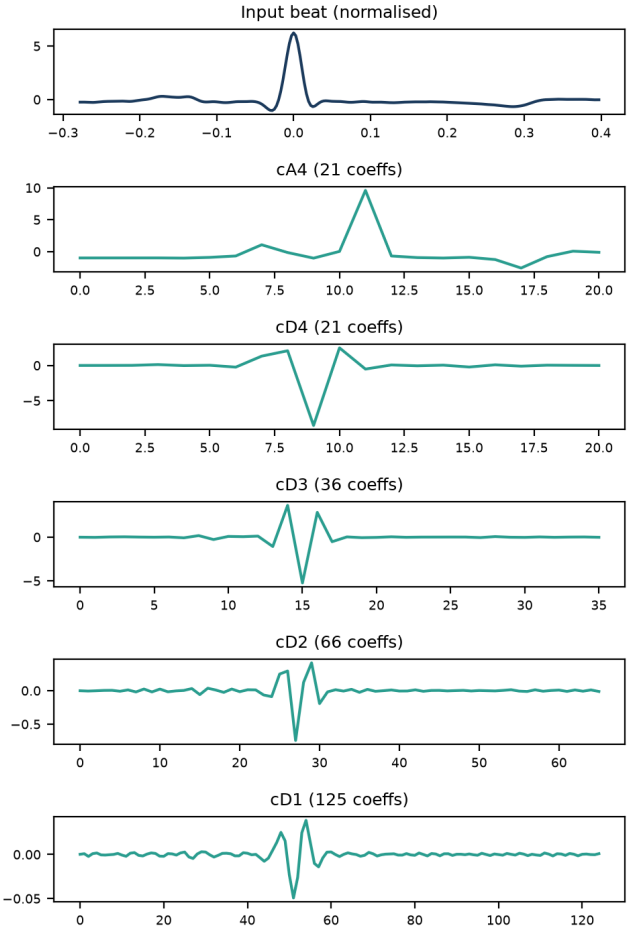


Fig. 3. Level-4 db4 wavelet decomposition of a real normal beat into one approximation band (cA_4) and four detail bands (cD_4–cD_1), the basis of the Approach 2 feature set.

extracts local morphological patterns; two convolutional blocks (16 then 32 filters, kernel sizes 7 and 5, each followed by ReLU and max-pooling) progressively summarise the waveform (Table IV, Fig. 4). The resulting sequence is passed to a bidirectional LSTM, which for each direction maintains

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \tag{17}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \quad \tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c), \tag{18}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \qquad h_t = o_t \odot \tanh(c_t), \tag{19}$$

and the forward/backward final hidden states are concatenated to capture temporal structure in both directions across the beat. A dropout layer ($p = 0.3$) regularises the 128-dimensional representation before a dense layer and softmax,

$$P(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{5} e^{z_j}}, \quad k \in \{N, S, V, F, Q\}. \tag{20}$$

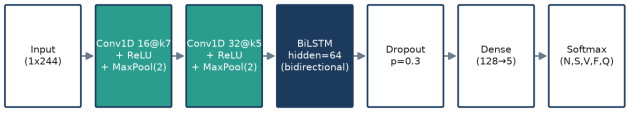


Fig. 4. 1D-CNN + BiLSTM architecture used in this study (matches the implemented model exactly; see Table IV).

TABLE IV
1D-CNN + BiLSTM Architecture (Approach 3)

Layer	Config.	Output shape
Input	raw normalised beat	$(1, 244)$
Conv1D + ReLU	16 filters, k=7, pad 3	$(16, 244)$
MaxPool1D	pool 2	$(16, 122)$
Conv1D + ReLU	32 filters, k=5, pad 2	$(32, 122)$
MaxPool1D	pool 2	$(32, 61)$
BiLSTM	hidden 64, bidirectional	$(61, 128)$
(last time step)	–	$(128,)$
Dropout	p=0.3	$(128,)$
Dense + Softmax	$128 \to 5$	$(5,)$

Trained with Adam (lr=10^{-3}), batch size 128, up to 25 epochs with early stopping (patience 5) on DS1 validation macro-F1.

The network is trained with Adam ($lr = 10^{-3}$), batch size 128, on the full DS1 set further split 85%/15% by record (GroupShuffleSplit) into training and validation beats, with up to 25 epochs and early stopping (patience 5) on validation macro-F1. An initial development run selected the checkpoint by validation loss with uncapped inverse-frequency class weights; because class Q has only single-digit support in DS1, its inverse-frequency weight became large enough to destabilise training (validation loss diverged while training loss fell, and DS2 recall for the majority class N collapsed to 0.38). This was corrected, and is reported rather than hidden, by (i) capping class weights at 10× and (ii) selecting the checkpoint by validation macro-F1 instead of validation loss, since loss computed over a handful of Q validation examples is too noisy to trust for model selection. All results in Section V use the corrected configuration.

E. Experimental Setup

Table V summarises the shared experimental configuration. All three models use the identical DS1/DS2 split and the identical preprocessed beats; hyperparameter selection for every model uses only DS1 (via grouped cross-validation or a grouped validation split); DS2 is touched exactly once, for the results reported in Section V. A fixed random seed (42) is used throughout for reproducibility. Training and evaluation ran on a standard CPU (no GPU was used or required at this model scale); wall-clock training time for the CNN-BiLSTM is not reported precisely, since development was performed on a shared machine

TABLE V
Experimental Configuration

Item	Value
Language / version	Python 3.12
Core libraries	wfdb, NumPy, SciPy, scikit-learn, PyWavelets, PyTorch, pandas
Random seed	42 (fixed across all models)
Train/test split	DS1 (22 rec.) / DS2 (22 rec.), patient-independent
Hyperparameter search	GroupKFold (k=5) within DS1 only
SVM training set	class-balanced DS1 subsample ($\leq$4,000/class)
RF, CNN-BiLSTM training set	full DS1 (51,001 beats)
CNN-BiLSTM val. split	15% of DS1 by record (GroupShuffleSplit)
Class imbalance handling	class_weight="balanced" (SVM, RF); capped inverse-frequency weights, max 10× (CNN)
Evaluation protocol	single evaluation on DS2, after hyperparameter selection

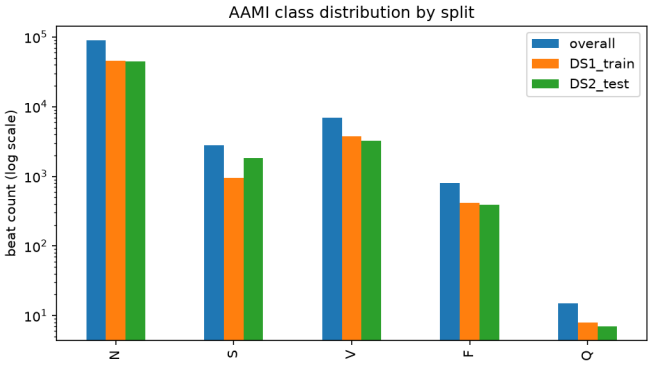


Fig. 5. AAMI class distribution (log scale) for the overall dataset, DS1, and DS2, confirming consistent, severe imbalance across the split.

under variable background load, but per-epoch compute cost was on the order of tens of seconds, consistent with the model's small parameter count.

V. Experimental Results and Analysis

A. Dataset Class Distribution

Fig. 5 and Table I show that class N accounts for roughly 89% of all beats in both DS1 and DS2, with S, V, F, and especially Q far rarer. This imbalance is the central reason accuracy alone cannot meaningfully rank the three approaches below: a model predicting N for every beat would already score close to 0.89 accuracy while being clinically useless.

B. Overall Performance Comparison

Table VI and Fig. 7 report accuracy, macro-F1, weighted-F1, and per-class precision/recall for all three models on the untouched DS2 test set. No hyperparameter in any model was selected using DS2.

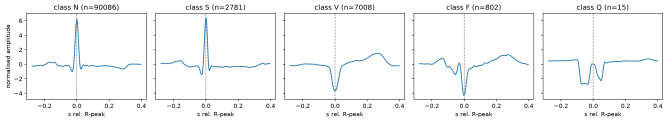


Fig. 6. Representative segmented, normalised beats for each AAMI class (record-level examples from the actual processed dataset).

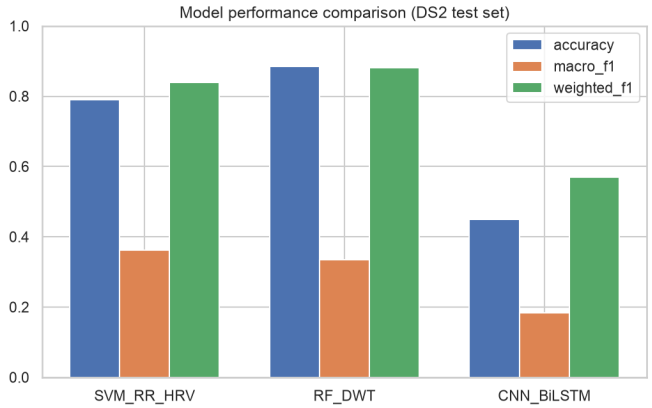


Fig. 7. Comparative model performance (accuracy, macro-F1, weighted-F1) on the DS2 test set.

Random Forest achieves the highest accuracy (0.884) and weighted-F1 (0.882); the SVM achieves the highest macro-F1 (0.362) and ROC-AUC (0.812); the CNN-BiLSTM, in this configuration, trails both on every aggregate metric. These three metrics do not agree on a single "best" model, which is itself the main empirical finding of this study and is discussed in Section VI.

C. Per-Class Analysis and Confusion Matrices

Fig. 8–10 show the DS2 confusion matrix for each model, and Fig. 11 compares per-class F1 directly.

Class N is handled well by all three (F1 0.884–0.937), except the CNN-BiLSTM (F1 0.621, recall 0.460), whose capped class weighting deliberately trades some N recall for minority-class recall. Class V is the best-handled minority class for every model (F1 0.724–0.757), consistent with ventricular ectopic beats having a relatively distinct waveform morphology that both wavelet statistics and learned convolutional filters can capture. Class S is poorly handled by all three models (F1 $\leq$ 0.065), indicating that none of the three feature representations used here cleanly separates supraventricular ectopic beats from normal beats – a known difficulty in the inter-patient literature, since S beats can closely resemble N beats morphologically and their timing signature is less distinctive than V's. Class F shows a striking divergence: the SVM achieves recall 0.943 (it rarely misses a true fusion beat) but precision only 0.053 (it also mislabels many non-F beats as F), while Random Forest and CNN-BiLSTM essentially never predict F correctly (F1 $\approx$ 0). This suggests RR timing alone is unusually (over-)sensitive

TABLE VI
Comparative Performance of the Three Approaches on the DS2 Test Set (49,691 beats)

				N		S		V		F		Q	
Model	Acc.	Macro-F1	W.-F1	P	R	P	R	P	R	P	R	P	R
SVM (RR/HRV)	0.789	0.362	0.839	0.967	0.814	0.111	0.046	0.682	0.851	0.053	0.943	0.001	0.143
Random Forest (DWT)	0.884	0.334	0.882	0.942	0.932	0.074	0.006	0.638	0.835	0.000	0.000	0.000	0.000
CNN-BiLSTM	0.449	0.183	0.570	0.957	0.460	0.037	0.127	0.151	0.539	0.000	0.010	0.000	0.000

P = precision, R = recall, W.-F1 = weighted-F1. ROC-AUC (macro, one-vs-rest): SVM 0.812, RF 0.730, CNN-BiLSTM 0.563. Q has only 7 test beats (Section III); its metrics are not statistically reliable.

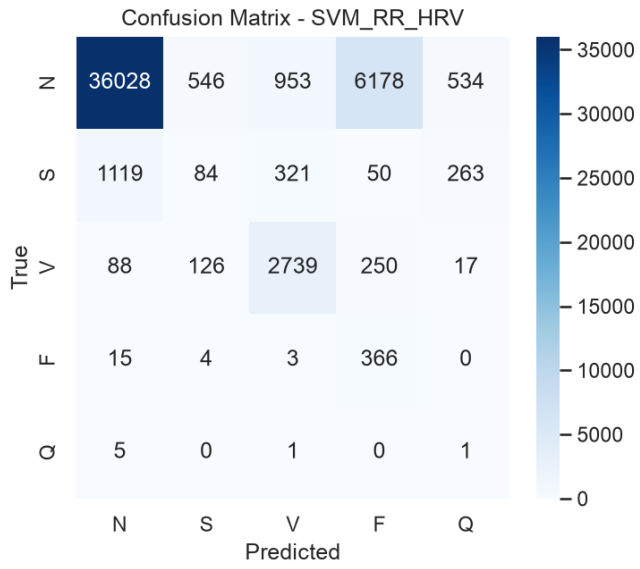


Fig. 8. Confusion matrix of the SVM (RR/HRV) classifier on DS2.

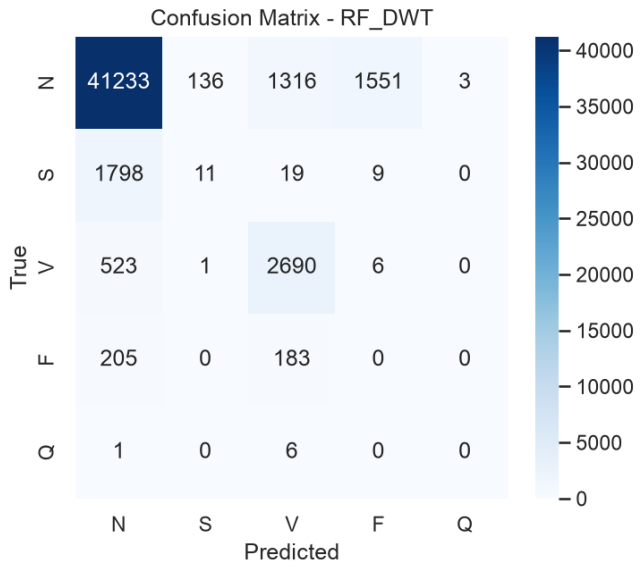


Fig. 9. Confusion matrix of the Random Forest (DWT) classifier on DS2.

to the fusion-beat pattern, while wavelet statistics and the compact learned representation do not isolate it. Class Q, with only 7 test beats, is not reliably measured by any model and is excluded from substantive comparison.

D. CNN-BiLSTM Training Behaviour

Fig. 12 shows training/validation loss and validation macro-F1 across epochs for the corrected configuration (Section IV-D). Training loss decreases steadily while validation loss is noisier but does not diverge, and validation macro-F1 plateaus around 0.29–0.31 before early stopping triggers at epoch 8, substantially lower than its performance on DS2 (macro-F1 0.183) – itself evidence of harder generalisation to genuinely unseen patients than to the held-out-by-record DS1 validation split, which is drawn from the same 22 training records' general population characteristics.

VI. Discussion

RR/HRV features (Approach 1). The strength of this approach is that RR timing is largely patient-invariant – a premature beat shortens the preceding interval similarly across individuals – which plausibly explains its best-in-class macro-F1 and ROC-AUC, and its unusually strong fusion-beat recall. Its weakness is equally clear: with no waveform information at all, it cannot distinguish classes that share similar timing but different morphology, visible in its very low F precision (many false fusion-beat predictions) and weak S recall.

Wavelet features (Approach 2). DWT statistics capture localized, multi-scale morphological structure and, combined with a Random Forest, give the best raw accuracy and weighted-F1 in this study – largely by being very good at the dominant class N. However, its near-zero recall on S and complete failure on F and Q, despite class-balanced training, suggest that a single beat's wavelet-coefficient statistics do not carry enough class-discriminative signal for these rarer categories once evaluated on truly unseen patients, even though in-sample (DS1 cross-validation) performance is comparable to the SVM's.

CNN-BiLSTM (Approach 3). The hybrid deep model underperforms both classical approaches here. We attribute this to the combination of (i) an intentionally

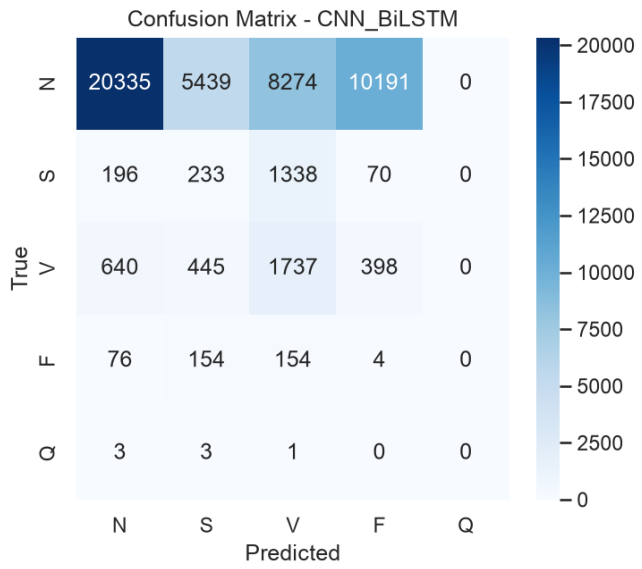


Fig. 10. Confusion matrix of the CNN-BiLSTM classifier on DS2.

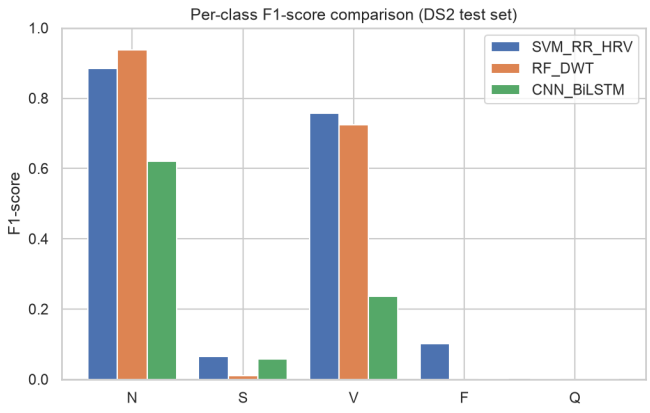


Fig. 11. Per-class F1-score comparison across the three models on DS2.

compact architecture and short training schedule within this project's compute budget, well below what Acharya et al. [4] and Liu et al. [5] used to report substantially higher accuracy; (ii) no explicit timing (RR-interval) input, so the network must infer temporal context from waveform shape alone; and (iii) the capped class-weighting scheme, which, while necessary to avoid the training instability documented in Section IV-D, further trades N recall for minority-class sensitivity that the model is not yet capable of exploiting well. This is a genuine limitation of the present configuration, not evidence that CNN-BiLSTM architectures are inherently weaker for this task.

Computational and practical implications. The SVM and Random Forest are each trained and evaluated in under a minute per run on a single CPU core once beats are cached, and require no specialised hardware. The

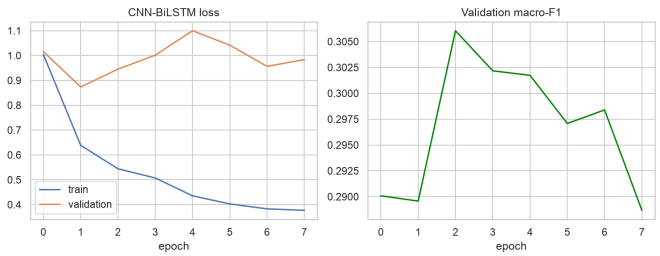


Fig. 12. CNN-BiLSTM training/validation loss and validation macro-F1 across training epochs (corrected class-weighting configuration).

CNN-BiLSTM, despite its small size, has a materially higher engineering and tuning cost (architecture search, class-weight tuning, longer training loops) for a worse outcome in this configuration – a practically relevant trade-off for a resource-constrained deployment, though it does not rule out a larger or better-tuned deep model eventually surpassing the classical approaches (Section VII).

Generalisation to unseen patients. The gap between DS1 cross-validation/validation performance and DS2 test performance, visible for all three models but most pronounced for the CNN-BiLSTM (Section V-D), underscores why the inter-patient split is not an optional methodological detail: it is the difference between an optimistic, patient-memorising estimate and a realistic one.

Limitations. This study uses a single lead (MLII), a 45-year-old research database recorded from 47 subjects, and does not evaluate on any second database; results should not be extrapolated to modern recording hardware or different patient populations. Class Q's single-digit support (Table I) means its reported metrics are illustrative only. Finally, this system is an offline, retrospective research prototype: it is not a clinically validated diagnostic or real-time monitoring tool, and none of the results above should be read as a clinical performance claim.

VII. Conclusion and Future Scope

This paper compared three heartbeat classification approaches – RR/HRV features with SVM, DWT features with Random Forest, and a 1D-CNN + BiLSTM trained on raw waveforms – under one identical, patient-independent evaluation protocol on the MIT-BIH Arrhythmia Database. No single model dominates every metric: Random Forest leads on accuracy and weighted-F1, the SVM leads on macro-F1 and ROC-AUC, and the compact CNN-BiLSTM trails both in this configuration. The most important conclusion is methodological rather than purely about which model "wins": under severe class imbalance and a genuine inter-patient evaluation, accuracy is an unreliable ranking criterion, and a model's apparent strength (e.g. the SVM's fusion-beat recall, or the Random Forest's overall accuracy) can mask a serious weakness (e.g. near-zero precision, or near-zero recall on

other minority classes) that only per-class, inter-patient evaluation reveals.

Future work should pursue: (i) larger and more diverse ECG datasets, and external validation on a second database (e.g. INCART or PTB-XL) to test cross-database, not just cross-patient, generalisation; (ii) improved minority-class detection, for example by combining RR-interval input with learned waveform features in a single model, or through imbalance-aware losses less prone to the instability documented here; (iii) a larger, longer-trained CNN-BiLSTM with a proper learning-rate schedule, to test whether the gap to classical methods closes or reverses with greater capacity; (iv) explainability methods (e.g. attention maps or SHAP values) to make per-class failure modes interpretable to a clinical reviewer; and (v) only after such validation, carefully controlled, prospectively validated real-time ECG monitoring studies – which remain well outside the scope of this retrospective, offline comparison.

Acknowledgment

The MIT-BIH Arrhythmia Database is provided by PhysioNet [8], originally compiled by the Beth Israel Hospital Arrhythmia Laboratory [7].

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